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PROJECT REPORT

ON

“Digital Contract Forensics System

Submitted To

Department of Cyber Security and Digital Forensics,
National Forensic Sciences University

For partial fulfilment of the award of the degree

MASTER OF SCIENCE

In

(DIGITAL FORENSICS AND INFORMATION SECURITY)

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DECLARATION

I hereby declare that the thesis entitled “Digital Contract Forensics System” is a research work done by me and no part of the thesis has been presented earlier and will be presented for any degree, diploma or similar title at any other institute/university.

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Pratap Mandal**

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M.Sc. Digital Forensics and Information Security

Year of Passing: 2027

Date:

Place: NFSU Delhi Campus



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I certify that

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- Whenever I have used materials (data, theoretical analysis, and text) from other sources, I have given due credit to them by citing them in the text of the dissertation and giving their details in the references.
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CERTIFICATE

This is to certify that the work contained in the project entitled “**Digital Contract Forensics System**”, submitted by **Mohendra Pratap Mandal** (Enroll. No.: 250246002042) for the award of the degree of **Master of Science in Digital forensics and Information Security** to the **National Forensic Sciences University, Delhi campus**, is a record of Bonafede's research works carried out by her under my direct supervision and guidance.

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**With Sincere Regards,
Mohendra Pratap Mandal**

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ABSTRACT

In the evolving landscape of the modern world, industrialization plays a fundamental role in shaping economies and societies. With industries at the front of progress, the integration of technologies has become important to update processes and maximize efficiency. And hence, ICS have emerged as the backbone of critical infrastructures, enabling management of crucial variables like pressure, air quality, temperature, etc.

The functioning of contract depends on the interconnection of cloud & blockchain components. This convergence has reformed industrial processes by the benefits of interconnectedness, automation, data-driven decision-making etc. By uniting two distinct domains, this integration has made control and monitoring more efficient, cost-effective, and scalable.

This connection of blockchain technologies and components in industries have its own consequences. The convergence of digital contract technologies introduces unique cybersecurity challenges that demand instant attention. The interconnection of traditionally isolated OT systems with the vastness of the internet and the complexities of IT networks creates larger attack surface for cyber threats. Attackers can quickly exploit vulnerability, posing a significant risk to the integrity, reliability, and safety of industrial processes and critical infrastructures.

Keywords: *ICS, SCADA, testbed, MitM attack, ARP spoofing, PLC, HMI, Digital Contract, hashing, cryptography.*

LIST OF ABBREVIATION

ABBREVIATION	DESCRIPTION
ARP	Address Resolution Protocol
CIA	Confidentiality, Integrity, Availability
DCS	Distributive Control System
DMZ	De-Militarized Zone
DID	Decentralized Identity Management
HMI	Human-Machine Interface
ICS	Industrial Control System
IIoT	Industrial Internet of Things
IP	Internet Protocol
API	Application Programming Interface
BFT	Byzantine Fault Tolerance

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Chapter 1 – Introduction

In the changing paradigm of the modern world, industrialization plays a paramount role in shaping economics and societies. As industries remain at the forefront of innovation and development, the adoption of technologies has become imperative to update systems and enhance efficiency. ICS have thus emerged as the backbone of critical infrastructure with effective control over critical parameters such as pressure, air quality, temperature etc.

The operation of contract depends upon interaction between cloud & blockchain components. Such a fusion of technologies has transformed industrial processes through improved connectivity, automation, data driven decisions, etc. Merging these two fields, the integration has made control and monitoring more efficient, economical and scalable.

Such a convergence between industrial domains & blockchain technologies has implications on systems. The integration of digital contract technologies into the industrial environment results in distinct cybersecurity challenges that require an urgent and immediate resolution. Merging traditionally separate OT systems with the expanse of the internet and intricacy of IT networks provides larger attack vectors that cyber criminals may readily exploit.

1.1. Overview to the Digital Contract Forensic System

This is precisely where the developed system comes into play. Using a simplified blockchain architecture along with a cryptography-based hashing method, a trustworthy digital contract and evidence forensics system is formed. When a digital contract or evidence file is uploaded into the system, a unique hash value is calculated for the file using the SHA-256 algorithm. This unique hash value acts like a digital finger-print of the file. The alteration of even a single bit of a file would change the calculated hash completely. This hash along with additional relevant information such as the file owner, timestamp and supporting evidence is recorded as a block into the blockchain.

Each block is cryptographically chained to the block preceding it, so that an altered file would not just change the hash of the current file but also the hash of the subsequent block, resulting in a chain reaction which changes all of the subsequent blocks. This would thereby make it easy to identify an alteration and ensure that no block has been tampered. For the implementation of this blockchain the values stored have been written in JSON format, the structure of the chain is easily managed in this manner as well as showing all that a chain can actually perform without a complex implementation.

Along with adding and retrieving files, a simple verification system can also be performed using this system where, when a file is passed again to be verified, a hash value is recalculated and compared with the hash value previously recorded on the chain. If a match is found then the file is deemed safe; if it is not found the file will either not be present in the chain or has been tampered with.

The commands that can be accessed using this command-line system are:

Add a new digital contract or evidence file to the system

Verify an existing digital contract or evidence file against the chain

View all the data on the blockchain

While the implemented blockchain is extremely simplistic (no actual mining or distributed consensus) it would illustrate that with the help of cryptography based hashing method and the chain-linking mechanism the data can be reliably and securely stored. This system shows just how useful blockchain technology can be in the real world such as digital evidence systems.

This system can improve trustworthiness in digital evidence handling by ensuring that the information that has been gathered for a crime is safe, verifiable and tamper-proof.

1.2. Overview to System

Ensuring the security and authenticity of digital evidence is extremely important in forensic investigations especially against the growing number of cyber crimes. Standard CoC models are vulnerable to various threats like data alteration, manipulation and are also not transparent to properly investigate into cyber crimes. In order to overcome these limitations of conventional models, we propose a novel secure chain of custody model for digital evidence based on the blockchain technology that provides an immutable, decentralized and transparent platform to manage evidence. Our proposed system employs smart contracts for automatic tracking of the digital evidence in cloud forensic, ransomware attacks, and investigation into growing complicated cyber threats such as IoT forensics. Our model incorporates strong cryptographic hash functions, efficient decentralized consensus mechanisms and real-time access logging to increase forensic reliability and cyber security framework. We analyze the various legal and regulatory concerns, ethical implications and future research trends such as implementation of the advanced AI based forensic analysis and use of post-quantum cryptography along with comparison with already existing forensic custody models. Our analysis indicates the efficiency, security and scalability of our proposed method and also helps in addressing various concerns about the security of digital evidence by emerging technologies along with a concrete real world forensic application.

Problems :

1. Address !=Identity

A wallet address is just a string.

Hard to link to real identity unless :

- KYC exchange data
- IP logs
- Off-Chain evidence

2. Pseudonymity & lack of identity attribution

Blockchain systems inherently operate on pseudonymous identifiers (wallets addresses)

Problem it creates:

- No directmapping between a blockchain record & a legal entity
- Attribution of offchain correlation (IP logs ,KYC records, device forensics)

3. Off-chain Data Dependency

The system stores only the cryptographic hash of the contract, while the actual document remains off-chain.

- Blockchain ensures data integrity.
- Loss or alteration of the original file makes reconstruction difficult.

4. Immutability

Blockchain follows an append- only data model .once data is written it cannot be modified

Digital signing of a contract

The method you use to sign a contract digitally depends on the specific tools and platforms you have access to. The following are the typical methods you might find used to sign contracts digitally:

Scan and upload: Hand-sign the contract on paper, scan the contract and then upload it as an image within your document.

Pros: Comfortable to use.

Cons: Very time-consuming and does not have security features.

Microsoft Word's signature line: Add a line in the document to have your digital signature there, then you can sign using this tool.

Pros: Has features built-in to Microsoft Office

Cons: Somewhat cumbersome to use and lacks advanced security features

Online signature tools: Upload documents into an online signature platform and sign electronically. Platforms like DocuSign or Adobe Sign are good examples of this.

Pros: Easy to use, secure.

Cons: Some versions of this tool lack the proper verification of user identity

CLM platform e-signatures: Use the features included within contract lifecycle management software like Concord to sign documents digitally.

Pros: Secure and compliant, the features fit in seamlessly with managing other contracts.

Cons: Usually only accessible with a subscription service to the CLM platform. CLM platforms such as Concord allow secure and compliant electronic signature creation, along with a range of editing and organizing tools to manage contracts.

1.3. Digital Contracts

A digital contract is a legally binding agreement executed via digital mediums. Also known as e-contracts, digital contracts are drafted, executed, stored, and managed in an electronic format. Digital contracts have the following characteristics:

Electronic format: the contract is produced in a digital format and does not necessitate the use of paper copies.

Legally binding: digital contracts are legally enforceable just like paper contracts if they meet all necessary legal criteria (e.g., offer, acceptance, consideration).

Electronic signatures: these serve the role of handwritten signatures by allowing users to seal their approval on the document digitally.

Accessible: cloud storage and the use of servers means that authorized users can access documents at any time and from any location.

Secure: advanced technology, such as encryption, is utilized to keep contracts safe from third parties and unauthorized access.

Timestamps: timestamps are often included to display an exact record of events and edits throughout the contract's lifecycle.

Interoperable: the contract can easily be integrated into other electronic systems used in business, for example accounting and project management.

Global reach: E-signature laws make digital contracts legally enforceable in a wide variety of locations and therefore they can be used for international dealings.

As we'll see in the following sections, these features help make digital contracts legal, while also making them more efficient and convenient than traditional paper agreements.

1.3. Vulnerabilities

Smart contracts are interesting to attackers, because they automatically perform actions, and usually operate on a large flow of data, and consume resources:

1. Reentrancy attacks

The existence of reentrancy attack vectors stems from the imperative nature of Solidity smart contract execution. Individual lines of code must execute completely before the next one can be initiated. Therefore, when a contract makes an external call to another contract, the executing contract must suspend until it gets back. This gives the called contract control over the next steps, and a possibility for an infinite loop:

3. Insecure randomness

Cryptographic algorithms rely on sources of [random numbers](#) when generating keys and performing other actions for smart contracts. A fundamental principle of cryptography is that random numbers must be unpredictable, and while that sounds obvious, it's often quite challenging to achieve. If the sources of randomness are not strong and there is any predictability in the values that they generate, this could provide several opportunities for attacks to circumvent the cryptographic protections that are fundamental to the integrity of smart contracts.

4. Gas griefing

The users have to spend gas to carry out a transaction or to call a smart

contract on the Ethereum platform. It is being spent to encourage miners to dedicate their computing power for validation of transactions. The gas prices depend on the availability of the gas, the demand and network size.

Gas griefing takes place when a user sends the gas amount necessary for executing the intended smart contract but less than needed to call another smart contract. If the smart contract doesn't check if there is enough gas to execute subcalls, it may fail to execute, thus impacting the application logic drastically.

5.Timestamp Dependence

The timestamp is created by the node which runs the smart contract. Since Ethereum is decentralized it's extremely hard to ensure that all the node clocks have the same time. In turn a node can control the timestamp it used to create a logic attack against any contract which depends on the `block.timestamp` variable for time-dependent functions.

These Modbus codes are sent to the PLC from an HMI to control the functioning in the industry.

1.4. Modbus Code Injection

Modbus code injection involves the injection of malicious code into a PLC that utilizes the Modbus protocol. This attack aims to compromise the integrity and security of the ICS system by manipulating the functionality and behavior of the PLC.

The consequences of Modbus code injection can be severe. By injecting malicious code into the PLC, an attacker can manipulate control processes, alter sensor readings, or issue unauthorized commands. This can lead to troublesome effects on industrial operations, such as incorrect process control, equipment malfunctions, or physical damage to machinery. Once the attacker gains control over the PLC, they can exploit it as a foothold to further compromise the entire ICS network, potentially leading to broader system breaches and more extensive damages.

1.5. Aim and Objective of the Project

The aim of this project is to analyze the effects of different attacks on an ICS by simulating these attacks on the GRFICS testbed. The primary focus is to understand the vulnerabilities and consequences of potential attacks in order to enhance the security of ICS environments.

The specific objectives of the project are as follows:

1. **Setting up the GRFICS testbed simulation environment and an Machine:**

This objective involves configuring the GRFICS testbed with the necessary hardware, software, and network components. Then configuring a Linux based machine into the environment to simulate the attacks.

2. **Perform a MitM attack on the GRFICS testbed using ARP Spoofing:**

The objective is to execute a MitM attack on the testbed, which involves ARP spoofing to intercept communication between devices within the ICS. By performing this attack, we aim to evaluate the potential consequences on system

integrity and confidentiality, thus gaining insights into the vulnerabilities of the system.

3. Perform Modbus codes injection attack on the PLC:

This objective focuses on injecting malicious Modbus codes into the PLC within the GRFICS testbed. By injecting malicious code, we aim to study the effects of such attack, including potential disruptions to control processes and compromise of critical system operations.

By achieving these objectives, this project will contribute to a better understanding of the effects of attacks on ICS.

1.6. Scope of the Project

The design and development of a system that uniquely identifies digital files using cryptographic hash (SHA-256) values.

The construction of a custom blockchain architecture, implemented in Python, which maintains file hashes along with metadata including the owner's identity, creation time and supporting evidence's details.

The provision of data immutability by chaining the blocks of the blockchain together through the hash of the previous block.

The development of a verification system where users can check a file's authenticity using its current hash against the blockchain record.

The facility for tamper-proof evidence, making changes to original files easily discoverable through their hash mismatches.

A timestamped and transparent ledger of all digital contracts and evidence submitted.

The implementation of a simplified lightweight blockchain system which does not depend on any outside infrastructure, hence suitable for academic and prototyping purposes.

A demonstration of blockchain's applications in digital forensics by its ability to keep and check sensitive digital files, specifically network capture files (pcap), integrity.

A straightforward command line interface to enable file additions, verifications, and retrieval of recorded blockchain entries.

Improvement of the basic concepts like cryptographic hash, data structure of blockchain and digital evidence management. ic concepts like cryptographic hash, data structure of blockchain and digital evidence management.

Chapter 2 – Literature Survey

AI enabled chain of custody Blockchain based chain of custody for digital Forensics integrity.

Dr. Paresh R Rathod
Shree Swaminarayan College of Computer Science
Sardarnagar, Bhavnagar, Gujarat, India
drpareshrathod@gmail.com

Proposed a blockchain framework using smart contracts for chain- of – custody (CoC) in cloud / IoT forensics . Focuses on tamper-proof evidence storage and ransomware tracking through immutability and access controls.

Blockchain based Digital Forensics Investigation Framework in the Internet of Things& Social System

Mr.S. Nelson M E., Assistant Professor,
Department of Information Technology
V. S. B. Engineering College Karur, Tamilnadu
Mr. S. Karuppusamy B.Tech., Department of Information Technology

Presented a decentralised Digital Forensic framework for IoT /Social environment .Ensures evidences provenance ,privacy across borders & proof – of – existance via blockchain distributied ledger.

Development of Blockchain based smart contracts for Digital Forensics

Mayuresh Manohar Gaikwad1
1Post Graduate Student, Amity Institute of Forensic Sciences
Amity University, Sector-125, Noida, U.P-201301, India

Presents a proposal to develop smart contracts tailored for digital forensics operations.Automates CoC, consensus mechanism & cryptographic privacy to prevent tampering & enhance evidence integriety.

A game theoretic study of attack and defense in cyber-physical systems

Ma, C. Y., Rao, N. S., & Yau, D. K. | 2011 IEEE Conference on Computer Communications Workshops

The authors present game theoretic formulations for both attack and defence aspects using different cost and benefit functions and budgets. By examining the results of these games, they demonstrate that the Nash Equilibrium determines the system's survival. They conclude by showing that budget limits play a significant role in the security of cyber-physical systems.

Smart Forensics: A Blockchain Contract Approach Review

Smit Bhanushali
Department of Computer Engineering,
Saffrony Institute of Technology,
Mehsana, Gujarat, India

The research paper focused on the integration of digital forensics & blockchain technology. Ensuring accurate provenance records is crucial for evidence credibility and integrity in digital forensics. Blockchains offer properties to address these needs, but tailored designs are necessary for efficient provenance extraction.

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Chapter 3 – Methodology

1. Introduction

The proposed system employs a forensics-driven, integrity-first approach integrating cryptographic hashing, a blockchain-like, append-only log and evidence correlation for digital contract validation. The system process will ensure:

- Integrity Check (Tamper Evidence)
- Traceability (Time-stamped entries)
- Reproducibility (Deterministic hashing)

2. System Architecture

The system comprises the following modules:

1. Input Acquisition Layer
2. Hash Generation Module
3. Blockchain Storage Layer
4. Evidence Logging Module
5. Verification Engine
6. User Interface Layer (Tkinter GUI)

Each module is self-sufficient and can operate separately, but in unison, they form a cohesive forensics pipeline.

3. Input Acquisition

The system accepts digital contracts in file format (.txt, .pdf).

Procedure:

The file is selected by the user through the GUI

The file is read in binary format to prevent any sort of encoding inconsistencies

This guarantees:

The bit-stream of the file remains the same, irrespective of the platform

4. Cryptographic Hash Generation

The core of the methodology is the use of **SHA-256**.

Steps:

1. Files are read in sections (size is 4096 bytes)
2. Each chunk is processed by the hash function.
3. This final digest is outputted as a hex string (always the same length)

Properties used:

Determinism-input in, same output always

Avalanche Effect-tiny change, huge difference in output

Collision Resistance-unlikely to find two different inputs with the same hash.

Outcome:

A **unique digital fingerprint** representing the contract.

5. Blockchain-Based Storage Mechanism

A lightweight blockchain structure is implemented to ensure **tamper-evident storage**.

Block Structure:

Each block contains:

- Index
- Timestamp
- Contract hash
- Owner identity (user input)
- Evidence metadata
- Previous block hash
- Current block hash

Chain Formation:

- Blocks are linked using **hash pointers**
- Any modification breaks the chain integrity

Ensures:

- **Immutability (logical, not distributed)**
- **Sequential traceability**

6. Evidence Logging and Correlation

The system incorporates a simplified **evidence logging mechanism**.

Inputs:

- Email timestamps
- User-provided logs
- Contextual metadata

Purpose:

- Create the linking of the intent with the timeframe
- Simulate the practical forensic linkage:
 - Contract creation
 - Communication events

7. Persistent Storage

All the data of the blockchain are saved on a json-based ledger.

Characteristics:

- Human-readable
- Structured storage
- Persistent across sessions

Enables:

- Easy auditing
- Data recovery
- Reusability in verification

8. Verification Engine

The verification module performs **integrity validation** of a given contract.

Process:

1. Recompute hash of the input file
2. Traverse blockchain ledger
3. Compute hash and check against saved hashes

Outcomes:

- MatchesContract exists, and its content has not been altered.
- No Match Contract exists, but the content was changed, or it was unregistered.

This is a **deterministic comparison model** with zero ambiguity.

9. Tamper Detection Mechanism

A tamper can be identified using the basic principle of hash mismatch.

Logic:

- Even a single-bit modification alters the hash completely
- Due to the avalanche property of SHA-256

Result:

- Any unwanted changes are detected as and when they happen
- Original file not required to be compared against.

10. User Interface Interaction

Using Tkinter we implement a GUI.

Features:

- File selector
- Input fields for Owner and Evidence
- Buttons

Enhances:

- Usability
- Demonstration clarity
- Real-time interaction

11. Execution Workflow

User Input → Hash Generation → Blockchain Storage → Evidence Logging
 ↓
 Verification Request → Hash Recalculation → Ledger Comparison → Result

12. Forensic Soundness

The methodology aligns with core forensic principles:

- **Integrity Preservation** → via hashing
- **Chain of Custody (basic simulation)** → via blockchain logs
- **Repeatability** → deterministic hash computation
- **Auditability** → transparent ledger structure

13. Design Considerations

- Lightweight implementation (no heavy frameworks)
- Offline operation (no network dependency)
- Modular design for extensibility

14. Limitations in Methodology

- No automated evidence acquisition
- No cryptographic signature validation
- Centralized trust model (no consensus layer)



Chapter 4 – Implementation

4.1. Hardware & Software Required:

4.1.1. Hardware:

1. 30 GB free Storage
2. 3.5 GB RAM
3. Quad Core Processor

4.1.2. Software:

1. Visual Studio
2. Linux Mint
3. Command Terminal



4.2. Mathematical Model:

1. Cryptographic Hash Function Definition

Let:

- $H(x)$ be a cryptographic hash function (e.g., SHA-256)
- x be the input data

Then:

$$h = H(x)$$

Where:

- h = fixed-length hash output
- x = arbitrary-length input

2. Block Representation

Each block B_i in the chain is defined as:

$$B_i = (D_i, T_i, E_i, h_{i-1})$$

Where:

- D_i = contract hash (data)
- T_i = timestamp
- E_i = evidence metadata
- h_{i-1} = hash of previous block

3. Hash Chaining Function

The hash of the current block is computed as:

$$h_i = H(D_i \parallel T_i \parallel E_i \parallel h_{i-1})$$

Where:

- $\parallel$ = concatenation operator
- h_i = hash of current block

4. Recursive Dependency

Each block depends on the previous block:

$$h_i = H(B_i \parallel h_{i-1})$$

Expanding recursively:

$$h_i = H(D_i \parallel T_i \parallel E_i \parallel H(D_{i-1} \parallel T_{i-1} \parallel E_{i-1} \parallel \dots))$$

👉 This creates a **chain of dependency across all previous blocks**

5. ⚠️ Tamper Propagation Model

If any block B_k is modified:

$$D'_k \neq D_k$$

Then:

$$h'_k = H(D'_k \parallel T_k \parallel E_k \parallel h_{k-1}) \neq h_k$$

This propagates:

$$h'_{k+1} \neq h_{k+1}, \quad h'_{k+2} \neq h_{k+2}, \dots$$

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6. 🔍 Integrity Verification Condition

A blockchain is valid if:

$$h_i = H(D_i \parallel T_i \parallel E_i \parallel h_{i-1}) \quad \forall i$$

👉 If any equality fails → **chain integrity is broken**

7. 🧠 Security Properties (Mathematical View)

✓ Preimage Resistance

Given h , infeasible to find x such that:

$$H(\downarrow) = h$$

✓ Collision Resistance

Hard to find:

$$x_1 \neq x_2 \text{ such that } H(x_1) = H(x_2)$$

✓ Avalanche Effect

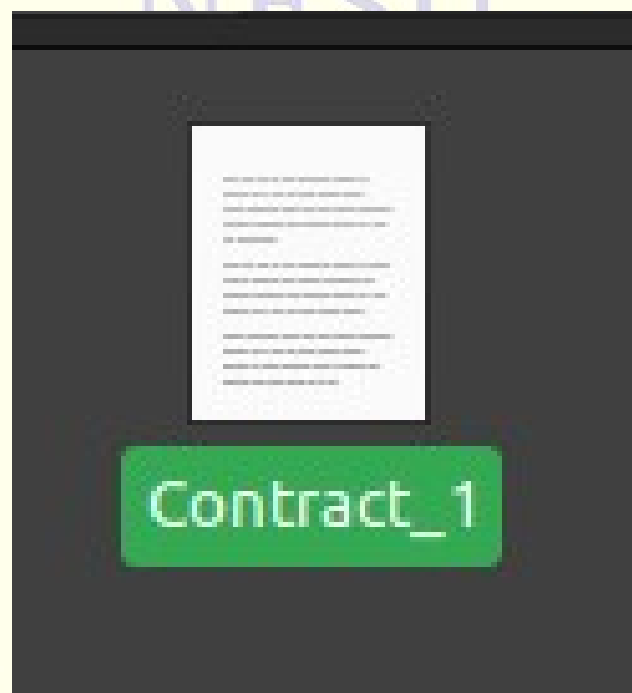
Small change in input:

$$x \rightarrow x'$$

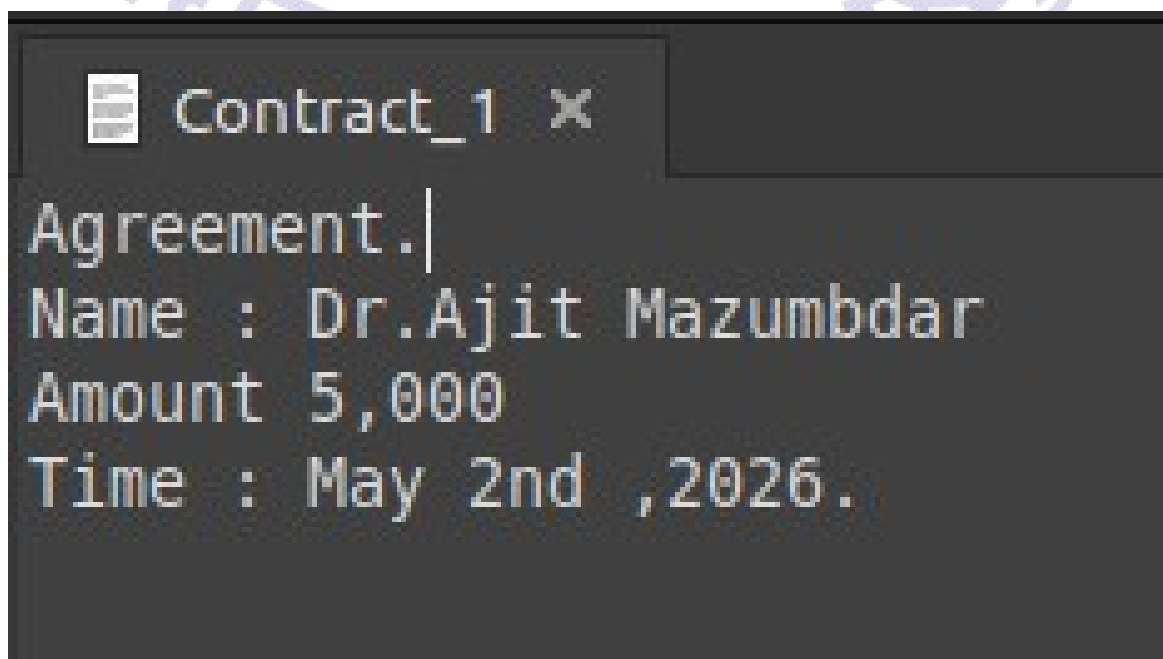
Produces:

$$H(x) \downarrow H(x')$$

Creating a new contract :



Contract_1.txt



Digital Contract Investigation :

On starting the program we get options :

```
===== Digital Contract Forensics System =====  
1. Add New Contract  
2. Verify Contract  
3. View Blockchain  
4. Exit  
Enter choice: █
```

Option 1.

We extractt the file path “/home/loon_dry/Desktop/NFS_LAB2/Contract_1

```
Name: Contract_1  
Type: Text (text/plain)  
Size: 72 bytes  
  
Location: /home/loon_dry/Desktop/NFS_LAB2  
Volume: unknown  
  
Accessed: Monday 04 May 2026 08:28:05 PM  
Created: Monday 04 May 2026 08:28:05 PM  
Modified: Monday 04 May 2026 08:28:05 PM
```

We enter the owners name & the email address

```
===== Digital Contract Forensics System =====
1. Add New Contract
2. Verify Contract
3. View Blockchain
4. Exit
Enter choice: 1
Enter contract file path: /home/loon_dry/Desktop/NFS_LAB2/Contract_1
Enter owner name: Mohendra Pratap Mandal
Enter evidence (email/time/log): mohendra@gmail.com

Hash Generated: 96afb53bf5f0bb965d7187f200a5ec6dc4b672af8db2102b0b811d51cb99961c
✅ Contract added to blockchain!
```

A hash value is generated which can be noted down.

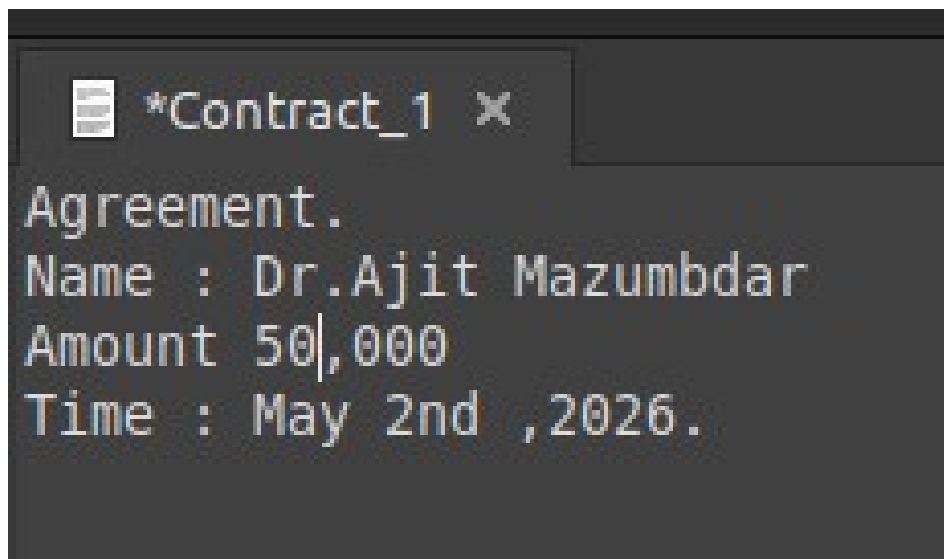
Option 2.

On successful establishment of a contract in our database repository.

We check for any tampering.

In our case we tamper it conditionally.

We change the value from 5,000 to 50,000



The file path is provided as in Option 1 to point out the same file.

Chapter 5 – Results and Findings

On comparing the hashes the result is known.

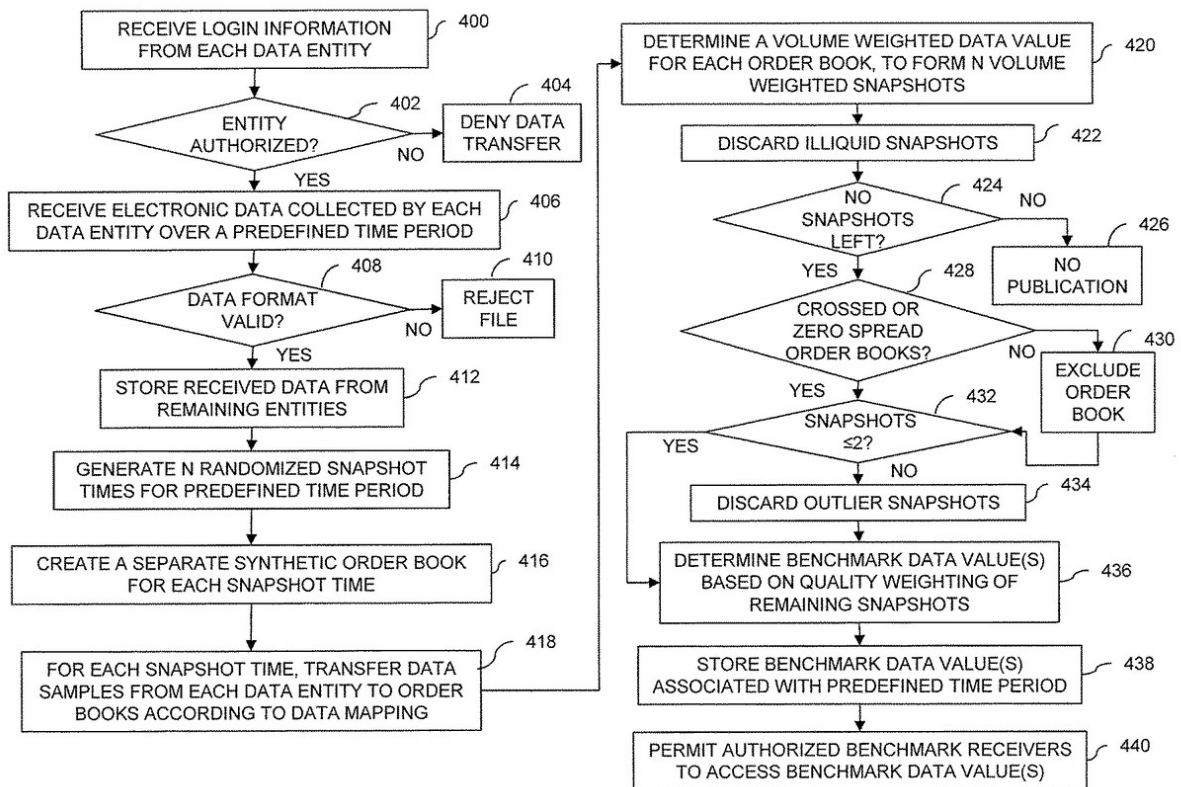
```
===== Digital Contract Forensics System =====
1. Add New Contract
2. Verify Contract
3. View Blockchain
4. Exit
Enter choice: 2
Enter contract file path to verify: /home/loon_dry/Desktop/NFS_LAB2/Contract_1

New Hash: 2f69d5e539de50790c47c9dc580f97426332e7f22bfd3d045e6f63df37d09209
⚠ Contract NOT FOUND or TAMPERED!
```

The hashes do not match ,if matched the result would be.

```
===== Digital Contract Forensics System =====
1. Add New Contract
2. Verify Contract
3. View Blockchain
4. Exit
Enter choice: 2
Enter contract file path to verify: /home/loon_dry/Desktop/NFS_LAB2/Contract_1

New Hash: 96afb53bf5f0bb965d7187f200a5ec6dc4b672af8db2102b0b811d51cb99961c
✅ Contract is VALID (Found in blockchain)
Owner: Mohendra Pratap Mandal
Timestamp: Mon May 4 20:35:12 2026
```



Chapter 6 – Conclusion

To summarize, this project illustrated how some aspects of blockchain and cryptography can be applied to digital contract forensics while revealing some of the intrinsic problems of such systems. As was shown through the hashing of the integrity of the contract, a small change made to the contract caused drastic changes to its cryptographic signature. This enables one to quickly see whether or not tampering has occurred; however, this one aspect doesn't allow us to trust the document entirely. For example, there is still the issue of verifying who added what part of the contract, as well as how to verify the meanings of each individual part of the contract (semantic analysis).

The method of hash chaining based on blockchain technology shows the concept of a tamper-evident system. Each transaction builds on the last so that any modification would cascade through the chain and become apparent. In our limited case of not having any consensus between many users/computers, it is clear how easily a central trusted authority can modify the data.

The other area of weakness shown by this project is reliance on off-chain data. The hash of the contract stored in the chain verifies that the document hasn't changed, but the document itself isn't stored nor is the semantic meaning verified. Additionally, manual evidence and chain-of-custody were a hindrance, so this system is likely to have some challenges regarding admissibility into court.

It seems that the best way to improve this system would be to introduce more pre-implementation features as well as more post-implementation features. Such additions as digitally signing each contract, incorporating a public key infrastructure (PKI), time stamping, using a true consensus among distributed computers would improve the pre-implementation aspect. Audit logging, monitoring, and possibly some sort of interface to other forensic tools could improve the post-implementation stage of this system.

Over all, one thing should be clear is that a combination of integrity mechanism of cryptographic and forensics methods should be employed. To be a complete system that actually has some legal merit one must extend the system to also provide identity binding and semantic verification along with consensus..

Chapter 7 – Future Scope

Future scope of institutional and technical advancement towards B2B and B2C authentication using blockchain: The system, in its present form, is providing a basis for integrity checking of digital contracts through the process of cryptographic hashing and the blockchain inspired ledger. To enable practical usage, the system could be developed to become a scalable, standards-compliant framework for authentication and trust suitable for both Business-to-Business (B2B) and Business-to-Consumer (B2C) applications. Below are the ways the system can be further expanded which will enable a technically rigorous and institutionally consistent road-map.

1. Decentralized Identity and trust anchors: Future system would use the DIDs (Decentralized Identities) and VCs (Verifiable Credentials) to tackle issues pertaining to identity attribution.

- * Use W3C standards for DIDs.
- * Tie the identity to public key cryptography.
- * Introduce trust anchors (such as regulators, certificate authorities) that will assure that entities are who they say they are.

Institutional impact:

- * Will support identity assurance over institutional boundaries.
- * Will help minimize dependence on central identity providers.
- * Will enable and uphold digital identity regulations.

2. Integration with Public Key Infrastructure (PKI): The system could be extended to involve PKI authentication and digital signatures.

- * Use asymmetric key-pairs for signing the contracts.
- * Verification through public keys/certificate chain.
- * Support X.509 certificates and enterprise grade identity systems.

Technical outcome:

- * Assures non-repudiation and authenticity.

- * Will enable legally verifiable digital contracts.
- * Comply with existing security standards in organizations.

3. Smart Contract based authentication and authorization: System logic of authentication can be encoded as a smart contract and deployed on a platform like Ethereum blockchain.

- * Access control rules in a smart contract.
- * Automation of verification and execution procedures.
- * Enforce conditional access to the digital asset/transaction.

Institutional impact:

- * Establish a trust-less, rule-based enforcement of access and transactions.
- * Minimize the need for manual work.
- * Increase transparency and audit trail.

4. B2B Integration: Inter-organizational trust framework: For B2B, it can be built into an infrastructure which would secure contract and identity exchange.

- * Cross-organization verification of digital contracts.
- * Blockchain based authentication of vendors and partners.
- * Traceability of supply chains through a cryptographic audit trail.

Use cases:

- * Procurement, vendor onboarding and contract management.
- * Regulatory compliance reporting.
- * Cross-border contractual agreements.

Outcome:

- * Strengthens inter-enterprise trust models.
- * Mitigates the risk of contract fraud and disputes.
- * Creates an auditable business transaction log.

5. B2C Authentication and digital interaction models: The system can provide user centric authentication mechanism that is both secure and transparent.

- * Password-less authentication with the use of keys.
- * Blockchain backed identity verification.
- * Consent based data sharing with the use of signed transactions.

Use cases:

- * E-commerce contract acceptance.
- * Digital subscription of services.
- * Secure user on-boarding.

Outcome:

- * User gets more control and privacy of their data.
- * Mitigate security issues related to passwords.
- * Increase consumer trust.

6. Secure Time-Stamping and Compliance with Regulation:

- * Trusted Timestamping Authorities (TSA).
- * Time anchoring based on blockchain networks.

Institutional Alignment:

- * Will prove legal admissibility of the digitally recorded assets/transactions.
- * Comply with legal framework such as IT Act in India, or similar policies.
- * Strengthen forensics and audit capabilities.

7. API-driven enterprise integration: System can provide RESTful APIs to integrate with the existing enterprise systems.

- * Enterprise resource planning (ERP)
- * Customer relationship management (CRM)
- * Identity and access management (IAM) solutions

Outcome:

- * Ease the integration into existing workflows.
- * Allow automation of contractual and authentication processes.

- * Ensure scalability over many organizations.

8. Scalability and Distributed Architecture: To avoid performance and scalability issues the proposed system can be modified to work in conjunction with:

- * Distributed blockchain network.
- * Layer 2 scaling solutions.
- * Off-chain storage coupled with on-chain hash anchoring.

Technical Outcome:

- * High transaction throughput and reduced latency.
- * Low operational cost.
- * System will scale for massive adoption.

9. Enhance Forensic and Audit Facilities: Further integration into forensic platforms might support in acquiring digital evidence automatically and then processing and analyzing evidence with tools to identify potential malicious attacks.

- * Automated evidence gathering and logging.
- * Digital forensic tools integration.
- * Advanced log analysis and anomaly detection.

Outcome:

- * Provide better forensics capability.
- * Increase the evidentiary value of the system logs.
- * Help in detecting new threats.

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